

Prasun Rimal

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Summary

Recent Computer Science and Mathematics graduate who builds and ships full-stack software across backend systems, AI workflows, and data-driven applications. Comfortable moving from algorithm design and statistical modeling to deployed production systems with real users. Proven ability to take projects from idea to live product — across trading platforms, AI operations tools, and graph optimization research — with clean architecture, precise engineering, and end-to-end ownership.

Education

St. Joseph's University, New York

Brooklyn, NY

B.S. Mathematics/Computer Science & B.S. Mathematics (Dual Degree)

May 2026

GPA: 3.811 / 4.000 | Honors: Magna Cum Laude, Sigma Iota Chi, Delta Epsilon Sigma, Dean's List (6 semesters), Dean's Scholarship

Relevant Coursework: Data Structures, Algorithm Design & Analysis, Software Engineering, Database Systems, Operating Systems, Data Science, Probability & Statistics, Linear Algebra

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, HTML, CSS

AI & Automation: LangGraph, Gemini API, pgvector, RAG, structured outputs, Pydantic, prompt engineering, output validation

Backend & APIs: FastAPI, Spring Boot, Spring Security, Spring Data JPA, REST APIs, JWT, SQLAlchemy, Pytest

Frontend: Next.js, React, TypeScript, Vite, Tailwind CSS, Recharts, Leaflet

Infrastructure: PostgreSQL, Docker, Git, GitHub Actions, Vercel, Render, Neon

Concepts: Graph algorithms, route optimization, OOP, design patterns, microservices, agile development

Projects

PayrollOps AI — Global Payroll Exception Operations Platform

[Live Demo](#) | [GitHub](#)

Next.js, TypeScript, FastAPI, LangGraph, Gemini, PostgreSQL, pgvector, Docker

- Built and deployed a full-stack internal operations tool that ingests synthetic global payroll CSVs, applies deterministic controls, and routes payment-impacting exceptions to a human review queue.
- Engineered a LangGraph retrieval-and-classification workflow grounding Gemini analyses in country-aware policy evidence, validating every response against a strict Pydantic schema with provider fallback.
- Implemented Argon2/JWT authentication, Admin/Reviewer authorization, confidence scores, citation visibility, and persistent audit logs for both operator and AI decisions.
- Deployed frontend on Vercel, FastAPI service on Render, and PostgreSQL/pgvector on Neon; added Pytest coverage for ingestion, retrieval, authorization, status transitions, and AI fallback behavior.

MockMarket — Full-Stack Paper Trading Platform

[Live Demo](#) | [GitHub](#)

Java 21, Spring Boot, React, TypeScript, PostgreSQL, JWT, Finnhub API

- Built and deployed a production-ready paper trading platform giving each registered user a simulated \$100,000 account for buying, selling, and tracking stocks with live Finnhub market quotes.
- Architected the backend as independently deployable microservices — auth, trading engine, market-data integration, and portfolio analytics — with protected REST APIs and environment-managed secrets.
- Implemented precise financial calculations with BigDecimal including cash balance, average cost tracking, holdings value, and gain/loss; deployed across Vercel, Render, and Neon with prod/dev separation.

Manhattan Last-Mile Delivery Optimizer — SURF Research Project

[GitHub](#)

Python, Flask, NetworkX, Pandas, Leaflet, OpenStreetMap

- Implemented Dijkstra and A* search on OSMnx street graphs to compare shortest-distance vs. delivery-aware routes enriched with traffic, curb-access, parking/loading, and NYC truck-route constraints.
- Designed a multi-objective cost function balancing distance, congestion, and route legality; demonstrated an optimized route that added only **35 meters** while reducing modeled delivery cost by **34.6 units** (34%).
- Developed an interactive Flask/Leaflet web app visualizing baseline and optimized routes; authored a formal research paper and delivered findings to a faculty and peer audience.

NYC Motor Vehicle Collision Analysis — Statistical Modeling & Crash Pattern Study

[GitHub](#)

R, tidyverse, ggplot2, statistical testing, logistic regression, spatial analysis

- Analyzed 187,082 NYC motor vehicle collision records from January 2024 to February 2026 to identify crash patterns, injury risk factors, and predictive signals.
- Built a reproducible R workflow for cleaning, feature engineering, borough-level analysis, time-of-day trends, injury/fatality rates, vehicle-type comparisons, and spatial coordinate coverage.
- Used chi-square tests to evaluate associations between injury outcomes and borough, contributing factor, and vehicle type.
- Trained and evaluated a logistic regression model for injury prediction, improving accuracy from about 55.5% to 63.16% with an AUC of 0.673.
- Produced publication-ready tables and visualizations for crash counts, injury rates, hourly trends, severity distribution, vehicle risk, ROC analysis, and spatial crash density.

Experience

St. Joseph's University, New York — Summer Undergraduate Research Fellowship

Brooklyn, NY

Summer Research Fellow

May 2025 – Aug. 2025

- Conducted faculty-advised research on graph-based optimization for urban last-mile delivery, iterating on algorithm design through a weekly agile review cycle and refining the approach based on structured feedback.
- Authored a formal research paper covering problem motivation, data sources, algorithmic approach, optimization framework, findings, and limitations; presented results to a stakeholder audience.

Lavner Education

Pleasantville, NY

STEM Instructor Intern

Jun. 2025 – Aug. 2025

- Taught project-based web development and introductory AI to students ages 8–14; adapted instruction across skill levels, debugged student code live, and explained complex technical concepts accessibly.

Centre for Health and Disease Studies — Liverpool School of Tropical Medicine—funded study Dang, Nepal

Field Research & Data Entry Assistant

Nov. 2020 – Dec. 2020

- Conducted structured household interviews across rural communities for a multi-week, Government of Nepal-approved field study on lymphatic filariasis.
- Collected and entered participant and blood-sample data into a shared research database, cross-checking entries for accuracy and consistency under field conditions.

Leadership & Awards

- **Vice President**, South Asian Student Association, St. Joseph's University New York (Aug. 2025 – May 2026) — led executive-board coordination, cultural programming, and campus outreach
- Magna Cum Laude | Sigma Iota Chi Honor Society | Delta Epsilon Sigma Honor Society
- Dean's Scholarship Recipient | Dean's List, 6 consecutive semesters | SURF Recipient